



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

Semester: SUMMER 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 02

Experiment Name: Introduction to Series-Parallel Circuits

Group No: 01

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Date of Submission: 29/06/2026

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Electrical Circuits I Laboratory EEE 101L

Department of Electrical & Electronic Engineering (EEE)

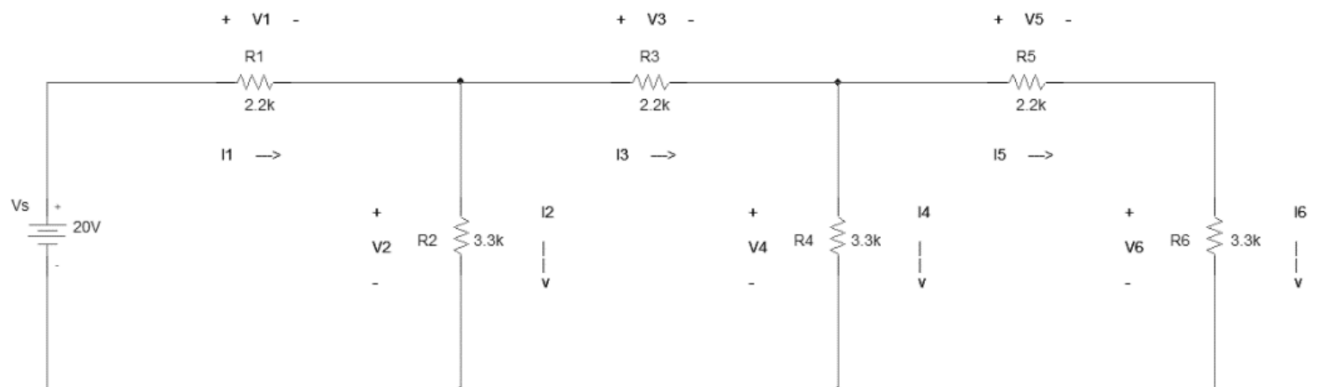
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Experiment No. 2

Introduction to Series-Parallel Circuits

1. Objective: The experiment is to acquaint the students with series-parallel circuits and to give them the idea about how to connect different circuits in bread board.

2. Theoretical Background: An electrical circuit is a continuous path or array of paths through which an electrical current can flow. The two different ways in which components of a circuit can be connected are called "series" and "parallel". In a series connection, components are connected one after another; therefore, the same current flows through all of them. In a parallel connection, the circuit components are connected side by side. That is, the positive and negative sides of each component are respectively connected together; therefore, each has the same potential drop across. In this lab, we will explore measurements of current and potential difference in simple circuits.



$$\text{Loop-1 KVL: } -V_s + V_1 + V_2 = 0 \rightarrow V_s = V_1 + V_2 \quad \dots\dots(i)$$

$$\text{Loop-2 KVL: } -V_2 + V_3 + V_4 = 0 \rightarrow V_2 = V_3 + V_4 \quad \dots\dots(ii)$$

$$\text{Loop-3 KVL: } -V_4 + V_5 + V_6 = 0 \rightarrow V_4 = V_5 + V_6 \quad \dots\dots(iii)$$

KCL,

$$I_1 = I_2 + I_3$$

$$I_3 = I_4 + I_5$$

3. Equipment:

- DC power supplies
- Resistors
- Bread board/ Trainer board
- One multimeter

4. Circuit Diagram:

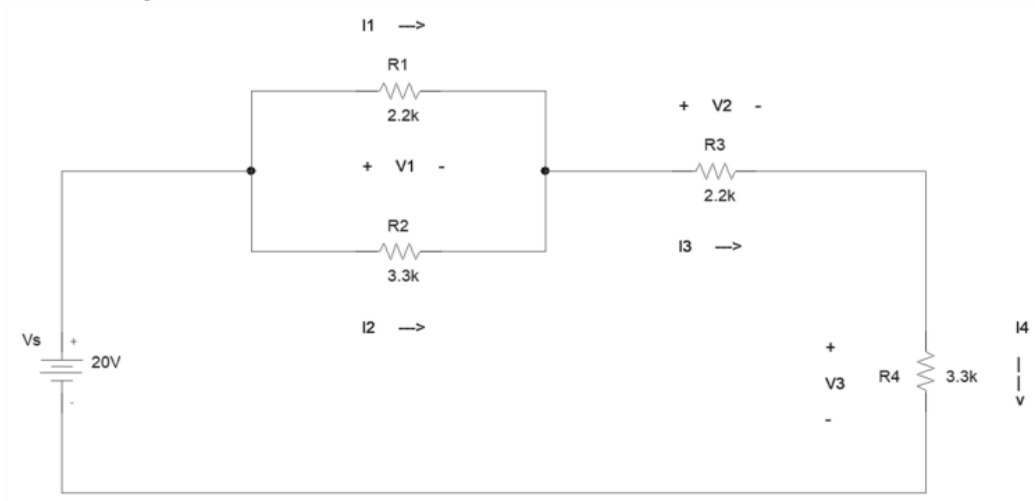


Fig. 1

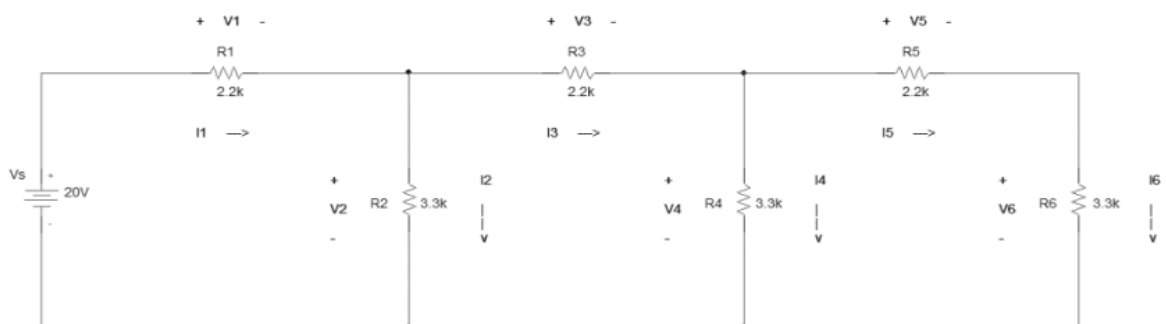


Fig. 2

5. Procedure:

- i. Set up the circuit as in the above figures.
- ii. Measure the voltage across the resistors and calculate the currents.

6. Data Table:

Fig. 2:

R1 (kΩ)	R2 (kΩ)	R3 (kΩ)	R4 (kΩ)	R5 (kΩ)	R6 (kΩ)
2.16	3.25	2.17	3.26	2.16	3.27

V₁ (V)	V₂ (V)	V₃ (V)	V₄ (V)	V₅ (V)	V₆ (V)
10.9	9.2	4.8	4.5	1.8	2.7

I₁ (mA)	I₂ (mA)	I₃ (mA)	I₄ (mA)	I₅ (mA)	I₆ (mA)
5.04	2.83	2.21	1.38	0.83	0.83

EEE101L
Exp-02

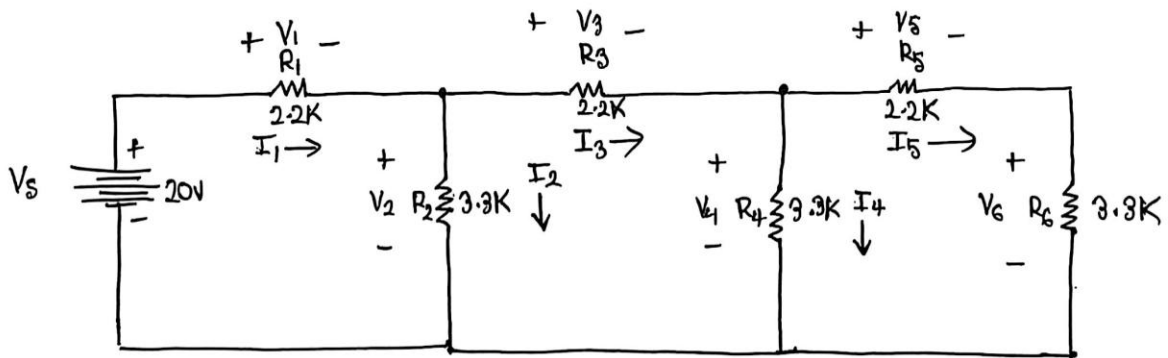
loop-1 KVL, $-V_3 + V_1 + V_2 = 0 \Rightarrow V_3 = V_1 + V_2$ — (i)

loop-2 KVL, $-V_2 + V_3 + V_4 = 0 \Rightarrow V_2 = V_3 + V_4$ — (ii)

loop-3 KVL, $-V_4 + V_5 + V_6 = 0 \Rightarrow V_4 = V_5 + V_6$ — (iii)

KCL, $I_1 = I_2 + I_3$

$I_3 = I_4 + I_5$



$V_s = 20.2$ loop-1 a, $V_3 = 20.2 = 10.9 + 9.2$ (20.1)

$V_1 = 10.9$
 $V_2 = 9.2$

$V_3 = 4.8$
 $V_4 = 4.5$ loop-2 a, $V_2 = 9.2 = 4.8 + 4.5$ (9.3)

$V_5 = 1.8$
 $V_6 = 2.7$ loop-3 a, $V_4 = 4.5 = 1.8 + 2.7$ (4.5)

$R_1 = 2.16K$	$I_1 = \frac{V_1}{R_1} = 5.04mA$	$I_1 = I_2 + I_3$ $I_3 = I_4 + I_5$
$R_2 = 3.28K$	$I_2 = \frac{V_2}{R_2} = 2.83mA$	
$R_3 = 2.17K$	$I_3 = \frac{V_3}{R_3} = 2.21mA$	
$R_4 = 3.26K$	$I_4 = \frac{V_4}{R_4} = 1.38mA$	
$R_5 = 2.16K$	$I_5 = \frac{V_5}{R_5} = 0.83mA$	
$R_6 = 3.27K$		

